



12628 - Simultaneous Confirmation and Spectral Characterization of a Giant Planet in Alpha Centauri A's Habitable Zone

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRI MRS				
	1	MRS Short Sequence	MIRI Medium Resolution Spectroscopy	(2) alf-Cen-Ab
	2	MRS Background Sequence	MIRI Medium Resolution Spectroscopy	(1) alf-Cen-Background

ABSTRACT

JWST Proposal 12628 (Created: Wednesday, May 20, 2026, 2:00:51PM Eastern Standard Time) - Overview

On August 11, 2024, in its most ambitious and technically demanding observation yet, JWST directly imaged the habitable zone of our nearest Sun-like star, Alpha Centauri A using the MIRI coronagraph at 15.5 μm . The significant effort dedicated to this high-risk, high-reward observation paid off and JWST/MIRI uncovered a ~ 225 K, 90-150 Earth mass candidate gas giant ~ 2 au from the star. Remarkably, the candidate is consistent in orbital and atmospheric properties with a source previously identified by VLT/NEAR observations at ~ 11.25 μm . However, two candidate detections do not equal a confirmed planet. Unambiguous confirmation requires a second JWST sighting. The orbital solutions converge and produce a strong prediction for the planet's expected location in August 2026, the only opportunity for robust confirmation. There is a strong urgency to these observations as a bright background star will prohibit any follow-ups until 2030 ($\sim$ Cycle 9).

We propose to leverage new technical breakthroughs in data analysis with MIRI MRS (demonstrated after Cycle 5 deadline) to spectroscopically confirm and characterize the planet for a fraction of the cost of the original discovery (10x reduction, and with much lower risk). We will detect NH_3 (16 sigma), expected in cold atmospheres, and measure $[\text{N}/\text{H}]$ to investigate the planet's formation location and/or the role of volatile delivery processes. This is one of only two, maybe three, planets on similar orbital scales as Jupiter where this is possible. We emphasize that ~ 11.5 - 13.5 μm moderate-resolution spectroscopy is not possible from the ground, even with the ELT, and JWST is the only telescope capable of making the above measurement.

OBSERVING DESCRIPTION

The observational landscape has dramatically shifted since the original detection was performed. A technical breakthrough in data analysis with MIRI MRS, achieved only recently after Cycle 5, has made it possible to simultaneously confirm and characterize the planet with the instrument. MRS observations dramatically reduce the risk by removing the need to perform a complex target acquisition process, leverage the fact that the planet and star are spectrally distinct to remain agnostic to speckles from both aCen A and B and background star KS5, and boost observational efficiency with a 10x reduction in clock time compared to the discovery program. We will leverage the molecular mapping strategy, specifically focusing on detecting the planet through its NH_3 features in Channel 3A.

The MIRI/MRS observations will position the expected position of the candidate at the center of the MRS FOV with the nominal pointing accuracy of JWST (0.1 arcsec, 1sigma) using astrometry for aCen A from Beichman et al. (2025) and the orbit predictions. We request one set of MRS sub-bands (1A, 2A, 3A). The NH_3 features are strongest in 3A and this will accomplish the goal of planet detection and characterization. We use 20 groups and 18 integrations to achieve a $\text{S}/\text{N} > 5$ at each wavelength, reaching up to $\text{S}/\text{N} \sim 50$ at the brightest wavelengths at the expected position of the planet. This S/N is expected to be achieved at separations outside ~ 1 arcsec and is required to achieve the spectral matching and extraction goals. While the core of the PSF will be saturated, no pixels will be partially/fully saturated at the expected planet position (from ETC calculations). A 4-

point dither pattern will mitigate the effects of IFU undersampling.

An observing window of ± 1 week around August 15, 2026 ensures the candidate's predicted position is well-constrained and that it is at a favorable separation for recovery. The exact date of the observation will be set in conjunction with the STScI planning team(s). Additionally, we will impose V3PA constraints to ensure that the planet does not fall in a PSF diffraction spike. Simulations show such configurations exist for the range of dates above. This is expected to improve the contrast sensitivity by at least an order of magnitude.

Proposal 12628 - Targets - Simultaneous Confirmation and Spectral Characterization of a Giant Planet in Alpha Centauri A's Habitable Zone

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	alf-Cen-Background	RA: 14 46 12.5300 (221.5522083d) Dec: -62 37 38.75 (-62.62743d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Parallax: 0.0" Epoch of Position: 2024.2	
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES				
	(2)	alf-Cen-Ab	RA: 14 39 22.5400 (219.8439167d) Dec: -60 49 52.62 (-60.83128d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0 Parallax: 0" Epoch of Position: 2000	
	Comments: Category=Unidentified Description=[Infrared sources] Extended=NO				

Proposal 12628 - Observation 1 - Simultaneous Confirmation and Spectral Characterization of a Giant Planet in Alpha Centauri A's Ha...

Observation	Proposal 12628, Observation 1: MRS Short Sequence												Wed May 20 19:00:51 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MRS Background Sequence (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	alf-Cen-Ab	RA: 14 39 22.5400 (219.8439167d) Dec: -60 49 52.62 (-60.83128d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0 Parallax: 0" Epoch of Position: 2000						
	Comments: Category=Unidentified Description=[Infrared sources] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction	
	F1000W	All MRS				NO			FULL			Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	20	18	1	Dither 1	4	72	4184.76	
	1	SHORT(A)	MRSSHORT		FASTR1	20	18	1	Dither 1	4	72	4184.76	

Proposal 12628 - Observation 1 - Simultaneous Confirmation and Spectral Characterization of a Giant Planet in Alpha Centauri A's Ha...

Special Requirements	Between Dates 10-AUG-2026:00:00:00 and 20-AUG-2026:00:00:00 Sequence Observations 1, 2, Non-interruptible
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Proposal 12628 - Observation 2 - Simultaneous Confirmation and Spectral Characterization of a Giant Planet in Alpha Centauri A's Ha...

Observation	Proposal 12628, Observation 2: MRS Background Sequence												Wed May 20 19:00:51 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [MRS Short Sequence (Obs 1)]												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	alf-Cen-Background		RA: 14 46 12.5300 (221.5522083d)			Proper Motion RA: 0 mas/yr						
				Dec: -62 37 38.75 (-62.62743d)			Proper Motion Dec: 0 mas/yr						
				Equinox: J2000			Parallax: 0.0"						
				Epoch of Position: 2024.2									
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS			NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type			Optimized For			Direction					
	1	4-Point			BACKGROUND			NEGATIVE					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	20	18	1	Dither 1	4	72	4184.76	
	1	SHORT(A)	MRSSHORT		FASTR1	20	18	1	Dither 1	4	72	4184.76	

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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